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Project Title:	Bacterial, Archaeal, & Fungal Diversity of the ISS		
Division Name:	Space Biology		
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Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
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Human Research Program Risks:	None		
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Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	None		
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Zip Code:	91109-8001	**Congressional District:**	27
Comments:			
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No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA ARC
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Flight Program:	ISS		
Flight Assignment:	NOTE: Title change in May 2015 to "Bacterial, Archaeal, & Fungal Diversity of the ISS--HEPA Filter System" from "Bacterial, Archaeal, & Fungal Diversity of Kibo-JEM Surfaces," per A. Chu/ NASA ARC via NASA JPL Rev. A to Task Order (Ed., 9/24/15) NOTE: Extended to 3/31/2018 per F. Hernandez/ARC (Ed., 8/31/17) NOTE: Extended to 6/30/2017 per A. Chu/ARC (Ed., 9/17/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Vaishampayan, Parag Ph.D. (Jet Propulsion Laboratory)		
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Task Description:	May 2015 Revision to Task Agreement (Ed. note: information received September 2015). Due to problems obtaining samples from Kibo-JEM, the project has been modified with a new title, “Bacterial, Archaeal, & Fungal Diversity of the ISS--HEPA Filter System.” The revised scope will allow the Jet Propulsion Laboratory (JPL) to undertake a 28-month flight study to examine the microorganisms associated with HEPA filters. The ISS-HEPA project will develop a comprehensive census of the microbial population on the HEPA filter collected from various locations of the International Space Station (ISS), using genetic biosignatures and computational modeling. This microbial census will provide NASA with the ability to accurately and confidently assess the status of microbes associated with closed habitation and crew health maintenance. In addition to overall microbial profiles, the JPL team will determine which microbial taxa pose particular threats to crew health. Original Task Description: Space microbiology research working groups in Japan, authorized by the Japan Aerospace Exploration Agency (JAXA), submitted proposals in response to the JAXA Announcement of Opportunity, 2012. The JAXA proposal that is managed by Principal Investigator Prof. Masao Nasu, Osaka University, contained several subgroups, including subgroup 4, headed by Dr. Takashi Yamazaki, JAXA. This subgroup 4 will conduct an investigation on “Analysis of microbial dynamics in the space habitation system by using new molecular microbial ecological techniques and also monitoring microbial contamination from air and surfaces of the ‘Kibo’ Japanese Experiment Module (Kibo-JEM).” The JAXA proposal submitted by subgroup 4 identifies the Jet Propulsion Laboratory (JPL) team headed by Dr. Kasthuri Venkateswaran as a collaborator to jointly explore the investigation. The Bacterial, Archaeal, and Fungal Diversity of Kibo-JEM Surfaces project (JPL-JAXA project) will undertake a three-year flight study to examine the changes in the microorganism population on the Kibo-JEM surfaces. The JPL-JAXA project proposes to develop a comprehensive census of the microbial population on the Kibo-JEM surfaces, using genetic biosignatures and computational modeling. This census will provide NASA and JAXA with the ability to accurately and confidently assess the status of microbes associated with closed habitation and crew health maintenance. In addition to overall microbial profiles, the JPL-JAXA team will determine which microbial taxa pose particular threats to crew health. Specific JPL-JAXA project goals will enable us to resolve applicable Human Research Program (HRP) integrated research plan (IRP) risks: (1) determine if changes in Kibo-JEM microbiome are occurring during a spaceflight mission and if so, (2) determine whether any changes noted would create a greater health risk for the crew. There are three aims proposed in this JPL-JAXA project and study of the Kibo-JEM microbiome. Through optimizations and comparative evaluations of methods designed to collect, process, analyze, and archive biomolecules sampled, the JPL-JAXA team will increase downstream sensitivity of laboratory assays as well as enhance the current understanding of the population dynamics of the Kibo-JEM module (Aim 1). The JPL-JAXA team will adapt molecular technologies currently in use with the National Institute of Health (NIH) – Human Microbiome Program (HMP) Roadmap—such as rRNA gene-variable region tag sequencing, whole genome shotgun sequencing (WGS), and DNA microarrays, as well as cloning and sequencing of 16S ribosomal DNA (rDNA)—to comprehensively elucidate the microbial diversity profiles associated with the Kibo-JEM surfaces (Aim 2). The final project deliverable to NASA will be the Kibo-JEM database, a compilation of all of the genomic sequences and genetic information of microbes encountered on and within the Kibo-JEM habitat (Aim 3). Computational modeling of the Kibo-JEM microbial population dynamics inferred from the database will then be carried out to predict the complete spectrum of organisms present. The JPL-JAXA project will compare microbial diversity, microbial characteristics, and functional capabilities of the microorganisms associated with the Kibo-JEM. The resulting database will enable HRP to evaluate the effect of spaceflight radiation and microgravity on the Kibo-JEM microbiome. The JPL-JAXA project in combination with results from studies of crew susceptibility will be used to determine the risk of microbiologically induced adverse health effects during a spaceflight mission. This assessment will allow HRP to determine if current operational and engineering controls used to mitigate microbiological risks are adequate or whether additional countermeasures should be developed.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The JPL-JAXA project proposes to develop a comprehensive census of the microbial population on the Kibo-JEM surfaces, using genetic biosignatures and computational modeling. This microbial census will provide NASA and JAXA with the ability to accurately and confidently assess the status of microbes associated with closed habitation and crew health maintenance. In addition to overall microbial profiles, the JPL-JAXA team will determine which microbial taxa pose particular threats to crew health.
	Microbiomes of the dust particles collected from the International Space Station and Spacecraft Assembly Facilities 1. The International Space Station (ISS) is a uniquely built environment due to the effects of microgravity, space radiation, elevated carbon dioxide levels, and especially continuous human habitation. Understanding the composition of the ISS microbial community will facilitate further development of safety and maintenance practices. 2. The primary goal of this study was to characterize the viable microbiome of the ISS-built environment. 3. A second objective was to determine if the built environments of Earth-based cleanrooms associated with space exploration are an appropriate model of the ISS environment. 4. Samples collected from the ISS (HEPA filter and vacuum debris) and dust particles from two cleanrooms at the Jet Propulsion Laboratory (JPL, Pasadena, CA) were analyzed by traditional cultivation, adenosine triphosphate (ATP), and propidium monoazide–quantitative polymerase chain reaction (PMA-qPCR) assays to estimate viable microbial populations. 5. The 16S rRNA gene Illumina iTag sequencing was used to elucidate microbial diversity and explore differences between ISS and cleanroom microbiomes. Statistical analyses showed that members of the phyla Actinobacteria, Firmicutes, and Proteobacteria were dominant in the samples examined but varied in abundance. 6. Actinobacteria were predominant in the ISS samples whereas Proteobacteria, least abundant in the ISS, dominated in the cleanroom samples. The viable bacterial populations seen by PMA treatment were greatly decreased. However, the treatment did not appear to have an effect on the bacterial composition (diversity) associated with each sampling site. 7. The results of this study provide strong evidence that specific human skin-associated microorganisms make a substantial contribution to the ISS microbiome, which is not the case in Earth-based cleanrooms. For example,

Task Progress:

Corynebacterium and Propionibacterium (Actinobacteria) but not Staphylococcus (Firmicutes) species are dominant on the ISS in terms of viable and total bacterial community composition. 8. The results obtained will facilitate future studies to determine how stable the ISS environment is over time. The present results also demonstrate the value of measuring viable cell diversity and population size at any sampling site. This information can be used to identify sites that can be targeted for more stringent cleaning. Finally, the results will allow comparisons with other built sites and facilitate future improvements on the ISS that will ensure astronaut health.

Characterization of Aspergillus fumigatus isolated from air and surfaces of the International Space Station

9. Two strains, ISSF 21 and IF1SW-F4, were isolated from the HEPA filter and the surface of the Cupola of the ISS, respectively. Using primers targeting the internal transcribed spacers ITS1 and 2, both isolates were identified as A. fumigatus. The whole genome sequence analysis of ISSF 21 revealed increased number of single nucleotide polymorphisms (SNPs) when compared to the reference A. fumigatus 293. SM profiles of both ISS isolates were compared to the reference (Af293) on 11 different media but no significant differences were observed. Finally, knowing that A. fumigatus is an opportunistic pathogen and microgravity highly influences the antibiotic susceptibility and pathogenicity of microorganisms, we examined pathogenicity of both ISS isolates using the zebrafish larval model. The ISSF 21 was more virulent than two clinical strains (Af293 and CEA10); the virulence of the second isolate IF1SW-F4 is being tested.

10. Proteomics and transcriptomics of these ISS isolates might reveal the molecular mechanisms of the increased virulence. Subsequently, if the enhanced virulence is attributed to the microgravity environment, NASA needs to develop countermeasures to protect the astronaut's health, whose immune system is reported to be compromised under microgravity.

Draft Genome Sequence of two Aspergillus fumigatus Strains Isolated from the International Space Station

11. In this study, we determined the draft genome sequences of two A. fumigatus strains (ISSFT-021 and IF1SW-F4).

12. Whole-genome shotgun sequencing was performed on an Illumina HiSeq2500 instrument with a paired-end module. A total of 27,987,752 and 10,752,032 paired end reads of 101-nucleotide were collected for ISSFT-021 and IF1SW-F4. The NGS QC Toolkit v2.3 was used to filter the data for high-quality (HQ) vector- and adaptor-free reads for genome assembly (cutoff read length for HQ, 80%; cutoff quality score, 20).

13. A total of 25,515,334 (~230x coverage) for ISSFT-021 and 8,466,430 (~75x coverage) for IF1SW-F4, high-quality vector-filtered reads were used for assembly with MaSuRCA (8) genome assembler (kmer size = 70).

14. The final assembly of the strain ISSFT-021 contains 312 contigs with a total size of 28,767,743 bp and an N50 contig length of 274.502 kb; the largest contigs assembled measures 813.103 kb and the GC% was 49.41.

15. Similarly, the final assembly of IF1SW-F4 strain contains 213 contigs with a total size of 28,249,183 bp and an N50 contig length of 367.421 kb; the largest contigs assembled measures 900.278 kb and the guanine-cytosine content% was 49.45.

16. Nucleotide sequence accession numbers: This whole-genome sequences of both the strains were deposited at DDBL/EMBL/GenBank under the accession no. LWRT00000000 and LWRU00000000. These A. fumigatus strains were also deposited in the USDA Agricultural Research Station and DSMZ public culture collections.

Metabolomic characterization of Aspergillus niger isolated from ISS.

17. This project involves the investigation of a fungal strain isolated from the International Space Station (ISS) US Lab surface, identified as Aspergillus niger.

18. The filamentous fungus A. niger is a workhorse for the biotech industry. It is used industrially as a production host for citric acid and enzymes. We hypothesized that because the ISS's microgravity and enhanced radiation conditions are potentially stressful for fungal growth, the ISS-isolated A. niger strain may produce novel secondary metabolites (SMs) that the citric acid-producing, genome sequenced A. niger strain (ATCC 11414) does not. To investigate this hypothesis, liquid chromatography-mass spectroscopy (LC-MS) was used to compare the SM profiles produced by these strains after growth in various conditions. 19. The data indicated that the ISS-isolated A. niger strain produces compounds not produced by the A. niger ATCC 11414 strain. In the interest of drug discovery and other industrial applications, we are determining the structures of these compounds. This process involves large-scale cultivation of the ISS-isolated A. niger strain, followed by extraction, purification, and characterization of each compound by nuclear magnetic resonance (NMR) spectroscopy.

20. Whole genome sequence of the ISS-isolated A. niger strain was carried out. In addition to providing a unique and novel drug discovery opportunity, this project provides insight into the metabolic phenotypes selected for environments of microgravity and enhanced radiation.

Molecular characterization of Bacillus issensis, sp. nov., isolated from various quarters of the International Space Station

21. As part of an ongoing effort to catalogue microbial communities inhabiting the International Space Station (ISS), crew-associated environmental samples were collected from the Japanese Kibo, Russian, and US research modules.

22. Initial analysis based on 16S rRNA gene sequencing identified 11 Bacillus isolates (two from Kibo Japanese Experiment Module (JEM), four from US Segment Harmony Node 2, and five from Russian Segment Zvezda module sites), all belonging to the Bacillus anthracis-B. cereus-B. thuringiensis group.

23. Isolates were further characterized by whole genome comparative analysis. Each isolate was sequenced, assembled, and aligned against all members of the B. cereus sensu lato group. Based on genome size estimates (5.2 – 5.3 Mbp), strain nucleotide identity (>99.99%), and maximum likelihood phylogenetic placement, all isolates were found to exhibit a very high level of similarity.

24. With respect to gene content, all isolates were inspected for cry proteins common to B. thuringiensis as well as toxins specific to B. cereus. No significant hits were found to any known cry genes, while full-length matches were found for multiple B. cereus toxin genes.

25. Finally, the isolates were closely screened against B. anthracis genomes, and no B. anthracis signatures were identified. Notably, all isolates contained the plcR ancestral “C” allele and lacked significant hits to pXO1 and pXO2 plasmids and toxins.

26. The collective results from the analysis confirm that these strains belong to a previously uncharacterized Bacillus species, hereafter referred to as Bacillus issensis. The type strain is ISSFR-003T (=NRRL B-65389T= DSMZ 101676T).

Solibacillus kalamii sp. nov., isolated from the International Space Station HEPA filter system

27. A Gram-positive, rod-shaped, endospore-forming, aerobic bacterial strain designated as ISSFR-015T, isolated from the International Space Station (ISS) high-efficiency particulate arrestance (HEPA) filter, was characterized by polyphasic taxonomy.

28. A comparative analysis of the 16S rRNA gene sequence (1,504 bp) of the strain ISSFR-015T showed the highest similarity to Bacillus isronensis MTCC 7902T (98.9%), followed by Solibacillus silvestris MTCC 10789T (98.5%), and Bacillus cecembensis MTCC 9127T (97.8%). DNA-DNA hybridization (DDH) analysis revealed the DNA dissociation values of strain ISSFR-015T, with other closely related species, to be in range 41–47%. The DNA G+C content of the strain ISSFR-015T was 45.4 mol%.

29. The major fatty acids were iso-C15:0 (45.2%) and C17:1omega10c (12.1%). The polar lipid profile contained diphosphatidylglycerol, phosphatidylglycerol, phosphatidylethanolamine, phosphatidylserine, and one unknown phospholipid. The isoprenoid quinones present in the strain ISSFR-015T were MK-7, MK-6, and MK-8. The peptidoglycan type of the cell wall was A4a L-Lys-D-Glu. Based on phylogenetic analysis, strain ISSFR-015T belongs to genus Solibacillus.

30. The polyphasic taxonomy data, including low DNA-DNA hybridization values and the chemotaxonomic analysis, confirmed that the strain ISSFR-015T represents a novel species, for which the name Solibacillus kalamii sp. nov. is proposed. The type strain is ISSFR-015T (=NRRL B-65388T= DSM 101595T).

Microbial succession in an Inflated Lunar/Mars Analog Habitat during a 30-day human occupation.

31. For potential future human missions to the Moon or Mars and sustained presence in the International Space Station, a safe enclosed habitat environment for astronauts is required. Potential microbial contamination of closed habitats presents a risk for crewmembers due to reduced human immune response during long-term confinement. To make future habitat designs safer for crewmembers, lessons learned from characterizing of analogous habitats is very critical. One of the key issues is that how human presence influences the accumulation of microorganisms in the closed habitat.

32. Molecular technologies, along with traditional microbiological methods, were utilized to catalog microbial succession during a 30-day human occupation of a simulated inflatable Lunar/Mars habitat. Surface samples were collected at different time points to capture the complete spectrum of viable and potential opportunistic pathogenic bacterial population. Traditional cultivation, propidium monoazide (PMA)–quantitative polymerase chain reaction (qPCR), and adenosine triphosphate (ATP) assays were employed to estimate the cultivable, viable, and metabolically active microbial population, respectively. Next-generation sequencing was used to elucidate the microbial dynamics and community profiles at different locations of the habitat during varying time points. Statistical analyses confirm that occupation time has a strong influence on bacterial community profiles. The Day-0 samples (before human occupation) have a very different microbial diversity compared to the later three time points. Members of Proteobacteria (esp. Oxalobacteraceae and Caulobacteraceae) and Firmicutes (esp. Bacillaceae) were most abundant before human occupation (Day-0), while other members of Firmicutes (Clostridiales) and Actinobacteria (esp. Corynebacteriaceae) were abundant during the 30-day occupation. Treatment of samples with PMA (a DNA intercalating dye for selective detection of viable microbial population) had a significant effect on the microbial diversity compared to non-PMA treated samples.

33. Statistical analyses revealed a significant difference in community structure of samples over time, particularly of the bacteriomes existing before human occupation of the habitat (Day-0 sampling) and after occupation (Day-13, Day-20, and Day-30 sampling). Actinobacteria (mainly Corynebacteriaceae) and Firmicutes (mainly Clostridiales Incertae Sedis XI and Staphylococcaceae) were shown to increase over the occupation time period. The results of this study revealed a strong relationship between human presence and succession of microbial diversity in a closed habitat. Consequently, it is necessary to develop methods and tools for effective maintenance of a closed system to enable safe human habitation in enclosed environments on Earth and beyond.

Publications in process:

Checinska, A., R. M. Kumar, D. Pal, S. Mayilraj, K. Venkateswaran. 2015. Solibacillus kalamii sp. nov., isolated from a high-efficiency particulate arrestance (HEPA) filter element used in the International Space Station. Submitted to IJSEM for a review.

Venkateswaran, K., A. Checinska, et al. 2016. Molecular characterization of Bacillus issensis, sp. nov., isolated from various quarters of the International Space Station. Submitted to Microbial Genomics for a review.

Bibliography Type:	Description: (Last Updated: 03/31/2026)
Abstracts for Journals and Proceedings	Romsdahl J, Blachowicz A, Chiang Y-M, Yaegashi J, Venkateswaran K, Wang C. "Multi-omic analysis of Aspergillus niger isolated from the International Space Station." 13th European Conference on Fungal Genetics. Paris, France, April 3-6, 2016. 13th European Conference on Fungal Genetics. Paris, France, April 3-6, 2016. Abstract Book, p. 574. , Apr-2016
Abstracts for Journals and Proceedings	Knox BP, Blachowicz A, Romsdahl J, Palmer JM, Huttenlocher A, Wang CCC, Keller NP, Venkateswaran K. "International Space Station Isolates of Aspergillus fumigatus." 7th Advances Against Aspergillosis Conference, Manchester, United Kingdom, March 3-5, 2016. 7th Advances Against Aspergillosis Conference, Manchester, United Kingdom, March 3-5, 2016. , Mar-2016

Abstracts for Journals and Proceedings	Romsdhal J, Blachowicz A, Chiang Y-M, Yaegashi J, Venkateswaran K, Wang CCC. "Metabolomic characterization of Aspergillus niger isolated from ISS." 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. , Nov-2015
Abstracts for Journals and Proceedings	Treangen T, Venkateswaran K "Identification and description of a novel Bacillus issensis, sp. Nov., isolated from the International Space Station." 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. , Nov-2015
Articles in Peer-reviewed Journals	Checinska A, Probst AJ, Vaishampayan P, White JR, Kumar D, Stepanov VG, Fox GE, Nilsson HR, Pierson DL, Perry J, Venkateswaran K. "Microbiomes of the dust particles collected from the International Space Station and Spacecraft Assembly Facilities." Microbiome. 2015 Oct 27;3:50. http://dx.doi.org/10.1186/s40168-015-0116-3 ; PubMed PMID: 26502721; PubMed Central PMCID: PMC4624184 , Oct-2015